

Hosp_MS.Sql

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1  -- =====
2  -- Hospital Management System - BY: Postgres.
3  -- =====
4  --Create Data base
5  CREATE DATABASE Hosp_MS;
6
7  --Creating the Tables
8  CREATE TABLE Department (
9      Dept_ID SERIAL PRIMARY KEY,
10     Dept_name VARCHAR(100) NOT NULL,
11     Location VARCHAR(255),
12     Num INT
13 );
14
15 CREATE TABLE Patient (
16     Patient_ID SERIAL PRIMARY KEY,
17     First_Name VARCHAR(50) NOT NULL,
18     Last_Name VARCHAR(50) NOT NULL,
19     Address TEXT,
20     Phone_Number VARCHAR(20),
21     Date_Of_Birth DATE
22 );
23
24 CREATE TABLE Users (
25     User_ID SERIAL PRIMARY KEY,
26     Name VARCHAR(100) NOT NULL,
27     Email VARCHAR(100) UNIQUE NOT NULL,
28     Password VARCHAR(255) NOT NULL,
29     Num VARCHAR(20),
30     Role VARCHAR(30) NOT NULL
31 );
32 );
33
34 CREATE TABLE Doctor (
35     Doctor_ID SERIAL PRIMARY KEY,
36     Name VARCHAR(100) NOT NULL,
37     Specialization VARCHAR(100),
38     Phone_number VARCHAR(20),
39     Dept_ID INT REFERENCES Department(Dept_ID) ON DELETE SET NULL,
40     User_ID INT REFERENCES Users(User_ID) ON DELETE CASCADE
41 );
42
43 CREATE TABLE Appointment (
44     Appointment_ID SERIAL PRIMARY KEY,
45     Date DATE NOT NULL,
46     Time TIME NOT NULL,
47     Patient_ID INT REFERENCES Patient(Patient_ID) ON DELETE CASCADE,
48     Doctor_ID INT REFERENCES Doctor(Doctor_ID) ON DELETE CASCADE
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49 );
50
51 CREATE TABLE Medical_Record (
52     Record_ID SERIAL PRIMARY KEY,
53     Diagnosis TEXT,
54     Treatment TEXT,
55     Record_Date DATE DEFAULT CURRENT_DATE,
56     Patient_ID INT REFERENCES Patient(Patient_ID) ON DELETE CASCADE,
57     Doctor_ID INT REFERENCES Doctor(Doctor_ID) ON DELETE SET NULL
58 );
59
60 CREATE TABLE Bill (
61     Bill_ID SERIAL PRIMARY KEY,
62     Total_Amount DECIMAL(10, 2) NOT NULL,
63     Payment_Data DATE DEFAULT CURRENT_DATE,
64     Patient_ID INT REFERENCES Patient(Patient_ID) ON DELETE CASCADE
65 );
66
67 CREATE TABLE Schedule (
68     Schedule_ID SERIAL PRIMARY KEY,
69     Available_Day VARCHAR(20),
70     Available_Time TIME,
71     Doctor_ID INT REFERENCES Doctor(Doctor_ID) ON DELETE CASCADE
72 );
73
74 --Insert into the Tables
75
76 INSERT INTO Department (Dept_name, Location, Num) VALUES
77 ('Cardiology', 'Building A - Floor 2', 15),
78 ('Neurology', 'Building B - Floor 1', 10),
79 ('Pediatrics', 'Building A - Floor 3', 20),
80 ('Orthopedics', 'Building C - Floor 1', 12),
81 ('Emergency', 'Ground Floor - Main Gate', 30);
82
83 INSERT INTO Patient (First_Name, Last_Name, Address, Phone_Number, Date_Of_Birth) VALUES
84 ('Mohamed', 'Abdelhameed', 'Cairo, Nasr City', '01011223344', '2000-01-01'),
85 ('Eman', 'Wally', 'Giza, Dokki', '01122334455', '1995-05-12'),
86 ('Rawan', 'Omar', 'Alexandria', '01233445566', '1988-10-20'),
87 ('Mostafa', 'Ahmed', 'Mansoura', '01544556677', '1992-03-15'),
88 ('Mostafa', 'Ibrahim', 'Tanta', '01099887766', '2005-07-08');
89
90 --Here Users with ROLE_ID[1:Admin, 2:Doctor, 3:Nurse, 4:Receptionist ]
91 INSERT INTO Users (Name, Email, Password, Num, Role) VALUES
92 ('Mohamed Amer', 'mo.amer@hospital.com', 'hashed_pass_123', 'U101', 'Admin'),
93 ('Dr. Ahmed Hassan', 'ahmed.h@hospital.com', 'doc_secure_456', 'U102', 'Doctor'),
94 ('Sara Mahmoud', 'sara.m@hospital.com', 'nurse_pass_789', 'U103', 'Nurse'),
95 ('Mona Ali', 'mona.a@hospital.com', 'recep_pass_000', 'U104', 'Receptionist'),
96 ('Dr. Omar Khaled', 'omar.k@hospital.com', 'doc_pass_321', 'U105', 'Doctor'),
97 ('Laila Ibrahim', 'laila.i@hospital.com', 'pass_nurse_202', 'U106', 'Nurse'),
98 ('Hany Salama', 'hany.s@hospital.com', 'recep_pass_303', 'U107', 'Receptionist'),

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99 ('Dr. Yasmin Ali', 'yasmin.a@hospital.com', 'doc_pass_404', 'U108', 'Doctor'),
100 ('Khaled Nour', 'khaled.n@hospital.com', 'admin_pass_505', 'U109', 'Admin'),
101 ('Dr. Tarek Hegazi', 'tarek.h@hospital.com', 'doc_pass_606', 'U110', 'Doctor'),
102 ('Noha El-Sayed', 'noha.s@hospital.com', 'nurse_pass_707', 'U111', 'Nurse'),
103 ('Mostafa Mahmoud', 'mostafa.m@hospital.com', 'recep_pass_808', 'U112', 'Receptionist'),
104 ('Dr. Reem Mansour', 'reem.m@hospital.com', 'doc_pass_909', 'U113', 'Doctor');
105
106 INSERT INTO Doctor (Name, Specialization, Phone_number, Dept_ID, User_ID) VALUES
107 ('Dr. Ahmed Hassan', 'Cardiology Specialist', '01122334455', 1, 2),
108 ('Dr. Omar Khaled', 'Neurologist', '01099887766', 2, 5),
109 ('Dr. Yasmin Ali', 'Pediatrician', '01211223344', 3, 8),
110 ('Dr. Tarek Hegazi', 'Orthopedic Surgeon', '01555667788', 4, 10),
111 ('Dr. Reem Mansour', 'Emergency Consultant', '01000112233', 5, 13);
112
113 INSERT INTO Appointment (Date, Time, Patient_ID, Doctor_ID) VALUES
114 ('2026-05-10', '10:00:00', 1, 1),
115 ('2026-05-10', '11:30:00', 2, 1),
116 ('2026-05-11', '09:00:00', 3, 2),
117 ('2026-05-11', '13:00:00', 4, 3),
118 ('2026-05-12', '15:00:00', 5, 4),
119 ('2026-05-12', '16:30:00', 1, 5),
120 ('2026-05-13', '08:00:00', 2, 2),
121 ('2026-05-13', '12:00:00', 3, 4);
122
123 INSERT INTO Medical_Record (Diagnosis, Treatment, Record_Date, Patient_ID) VALUES
124 ('Hypertension', 'Lisinopril 10mg once daily', '2026-05-10', 1),
125 ('Common Cold', 'Rest and hydration, Paracetamol as needed', '2026-05-10', 2),
126 ('Migraine', 'Sumatriptan and dark room rest', '2026-05-11', 3),
127 ('Seasonal Allergies', 'Antihistamine (Claritin) 10mg', '2026-05-11', 4),
128 ('Ankle Sprain', 'RICE protocol (Rest, Ice, Compression, Elevation)', '2026-05-12', 5),
129 ('Chest Pain (Stable)', 'Aspirin and referral for Stress Test', '2026-05-12', 1);
130
131 INSERT INTO Bill (Total_Amount, Payment_Data, Patient_ID) VALUES
132 (500.00, '2026-05-10', 1),
133 (150.50, '2026-05-10', 2),
134 (1200.00, '2026-05-11', 3),
135 (300.00, '2026-05-11', 4),
136 (250.75, '2026-05-12', 5),
137 (450.00, '2026-05-12', 1);
138
139 INSERT INTO Schedule (Available_Day, Available_Time, Doctor_ID) VALUES
140 ('Monday', '09:00:00', 1),
141 ('Monday', '14:00:00', 2),
142 ('Tuesday', '10:00:00', 3),
143 ('Wednesday', '09:00:00', 1),
144 ('Thursday', '13:00:00', 4),
145 ('Friday', '08:00:00', 5),
146 ('Saturday', '11:00:00', 3),
147 ('Sunday', '15:00:00', 2);
148

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149 --*****Queries*****
150
151 -- =====
152 -- 1. DATA MANIPULATION & UPDATES (DML)
153 -- =====
154
155 -- Add Gender column in patient table
156 ALTER TABLE Patient ADD COLUMN Gender VARCHAR(10);
157
158 -- Update Mohamed's Address
159 UPDATE Patient
160 SET Address = 'New Cairo, Egypt'
161 WHERE Patient_ID = 1;
162
163 -- Delete Mostafa from patient table (مطابق للـداتا الممسوحة ID تأكد أن الـ)
164 DELETE FROM Patient WHERE Patient_ID = 5;
165
166 -- =====
167 -- 2. DATA INTEGRITY & CHECK CONSTRAINTS
168 -- =====
169
170 -- Users تحديد الأدوار المسموح بها في السيستم ومنع أي قيم عشوائية جوه جدول
171 ALTER TABLE Users ADD CONSTRAINT check_allowed_roles
172 CHECK (Role IN ('Admin', 'Doctor', 'Nurse', 'Receptionist'));
173
174 -- التأكد من أن قيمة الفاتورة موجبة
175 ALTER TABLE Bill ADD CONSTRAINT check_positive_bill_amount
176 CHECK (Total_Amount >= 0);
177
178 -- التأكد من صيغة الإيميل الصحيحة
179 ALTER TABLE Users ADD CONSTRAINT check_user_email_format
180 CHECK (Email LIKE '%@%.%');
181
182 -- التأكد من أن تاريخ الميلاد منطقي
183 ALTER TABLE Patient ADD CONSTRAINT check_patient_dob
184 CHECK (Date_Of_Birth <= CURRENT_DATE);
185
186 -- التأكد من طول رقم التليفون
187 ALTER TABLE Doctor ADD CONSTRAINT check_doctor_phone_length
188 CHECK (LENGTH(Phone_number) >= 11);
189
190 -- Schedule التأكد من صحة أيام العمل المكتوبة في الـ
191 ALTER TABLE Schedule
192 ADD CONSTRAINT check_work_days
193 CHECK (Available_Day IN ('Saturday', 'Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday',
194 'Friday'));
195
196 -- =====
197 -- 3. VIEWS & DATA ANALYSIS

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198 -- =====
199
200 -- Payment_Date) تم تعديل اسم العمود لـ) الخاص بفواتير المرضى View إنشاء الـ
201 CREATE VIEW Patient_Billing_Summary AS
202 SELECT
203     p.Patient_ID,
204     p.First_Name || ' ' || p.Last_Name AS Patient_Name,
205     COUNT(b.Bill_ID) AS Total_Bills,
206     SUM(b.Total_Amount) AS Total_Paid
207 FROM Patient p
208 LEFT JOIN Bill b ON p.Patient_ID = b.Patient_ID
209 GROUP BY p.Patient_ID, p.First_Name, p.Last_Name;
210
211
212 -- =====
213 -- 4. ADVANCED QUERIES & REPORTING
214 -- =====
215
216 -- (JOIN) استعلام لمعرفة بيانات المريض والطبيب والقسم والموعـ
217 SELECT
218     p.First_Name || ' ' || p.Last_Name AS Patient,
219     d.Name AS Doctor,
220     dept.Dept_name AS Department,
221     a.Date AS Appointment_Date
222 FROM Appointment a
223 JOIN Patient p ON a.Patient_ID = p.Patient_ID
224 JOIN Doctor d ON a.Doctor_ID = d.Doctor_ID
225 JOIN Department dept ON d.Dept_ID = dept.Dept_ID;
226
227 -- Moh استعلام للبحث عن المرضى الذين يبدأ اسمهم بـ
228 SELECT * FROM Patient
229 WHERE First_Name LIKE 'Moh%';
230
231 -- (HAVING) استعلام عن المرضى الذين أنفقوا أكثر من 500
232 SELECT
233     p.First_Name || ' ' || p.Last_Name AS Patient_Name,
234     SUM(b.Total_Amount) AS Total_Spent
235 FROM Patient p
236 JOIN Bill b ON p.Patient_ID = b.Patient_ID
237 GROUP BY p.Patient_ID, p.First_Name, p.Last_Name
238 HAVING SUM(b.Total_Amount) > 500;
239
240 -- (ORDER BY & LIMIT) استعلام لمعرفة الطبيب الأكثر حجوزات
241 SELECT
242     d.Name AS Doctor_Name,
243     COUNT(a.Appointment_ID) AS Appointment_Count
244 FROM Doctor d
245 JOIN Appointment a ON d.Doctor_ID = a.Doctor_ID
246 GROUP BY d.Doctor_ID, d.Name
247 ORDER BY Appointment_Count DESC

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248 LIMIT 1;
249
250 -- استعلام فرعي للمرضى الذين حجزوا في قسم القلب (Subquery)
251 SELECT First_Name, Last_Name, Phone_Number
252 FROM Patient
253 WHERE Patient_ID IN (
254     SELECT Patient_ID
255     FROM Appointment
256     WHERE Doctor_ID IN (
257         SELECT Doctor_ID
258         FROM Doctor
259         WHERE Specialization = 'Cardiology Specialist'
260     )
261 );
262
```